

TraceX UIUX

UI/UX Document

TraceX — AI-Powered Criminal Network Analysis & Investigation Intelligence Platform

Version: 1.0 **Reference Documents:** TraceX PRD v1.1, SRS v1.0, Architecture v1.0

1. Purpose & Scope

This document defines the user experience for TraceX: navigation, screens, flows, components, states, and visual system. It is implementation-ready for a React + Cytoscape.js frontend (per the Architecture document) and is scoped to the four user roles defined in the SRS: Investigator, Intelligence Analyst, Supervisory Officer, System Administrator.

Non-negotiable UX constraint (from SRS FR-21 / BR-1): every entity, relationship, and AI-generated insight shown anywhere in the UI must visibly display its provenance (source-backed vs synthetic) and confidence level. This is not a details-panel afterthought — it shapes how components are designed from the start.

2. Design Principles

- Clarity over decoration.** This is an investigative tool used under time pressure — every screen should reduce cognitive load, not add visual noise.
- Evidentiary rigor is visible, not hidden.** Confidence level and source provenance appear on the graph itself (node styling), not only when a user clicks in — trust has to be legible at a glance.
- Nothing is presented as confirmed fact by default.** AI-suggested matches, key individuals, and anomalies are visually distinguished from investigator-confirmed findings (per BR-3) — e.g., dashed vs solid styling, “Suggested” vs “Confirmed” labels.
- Progressive disclosure for complexity.** A criminal network graph can get visually overwhelming fast — default views are simplified (e.g., 1–2 hop neighborhoods), with explicit user action required to expand further.
- Serious, restrained tone.** No playful illustration, no gamification. Professional, government-context appropriate visual language throughout.
- Never require memorizing state.** Every view a user can reach should be reachable again via clear navigation — no dead ends, no “how did I get here.”

3. User Roles & Primary Journeys

Role	Primary Journey
Investigator	Log in → open assigned case → explore connected network → confirm/reject flagged entity matches → trace a relationship path → export/reference findings
Intelligence Analyst	Log in → open cross-case analytics view → review key individuals / clusters / anomalies → drill into a specific network
Supervisory Officer	Log in → review case summaries → check unresolved entity-resolution queue status → view audit log

System Administrator	Log in → manage users/roles → monitor system health → configure data source connections
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4. Information Architecture / Navigation

Login

└─ Dashboard (role-dependent landing)
└─ Case Search / Case List
└─ Case Detail
└─ Connected Network Graph View
└─ Entity Detail Panel (provenance + confidence)
└─ Relationship Path Explorer
└─ Event Timeline
└─ Entity Resolution Queue (pending matches for this case)
└─ Cross-Case Analytics (Analyst+)
└─ Key Individuals
└─ Clusters / Communities
└─ Unusual Patterns
└─ Audit Log (Supervisory Officer+)
└─ Admin Panel (Administrator only)
└─ User & Role Management
└─ Data Source Configuration

Top-level navigation (persistent): Dashboard | Cases | Analytics (role-gated) | Audit Log (role-gated) | Admin (role-gated) | Profile/Logout.

5. Screens

Screen	Purpose	Primary Role(s)
Login	Authenticate	All
Dashboard	Role-appropriate landing summary (recent cases, pending reviews, alerts)	All
Case Search / List	Find and open a case	Investigator, Analyst, Supervisor
Case Detail	Container for graph, timeline, and resolution queue for one case	Investigator, Analyst, Supervisor
Connected Network Graph View	Visual exploration of entities/relationships (Cytoscape.js canvas)	Investigator, Analyst
Entity Detail Panel	Shows one entity's attributes, provenance, confidence, and connections	Investigator, Analyst
Relationship Path Explorer	Select 2 entities, view connecting path(s)	Investigator, Analyst
Event Timeline	Chronological view of events for a case/entity	Investigator, Analyst
Entity Resolution Queue	List of pending matches requiring confirm/reject	Investigator
Cross-Case Analytics	Key individuals, clusters, anomaly flags across cases	Analyst, Supervisor (view-only)
Audit Log	Searchable, read-only log of user actions	Supervisor, Admin

Admin Panel	User/role management, data source config	Admin
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6. Key User Flows

6.1 Confirm/Reject an Entity Match (core flow — realizes SRS FR-10, BR-3)

- Investigator opens **Entity Resolution Queue** from Case Detail.
- Sees a list of proposed matches, each showing both candidate entities side by side, their overlapping attributes (name/phone/address), and the system's confidence score.
- Investigator selects **Confirm** or **Reject**.
- On Confirm: entities merge in the graph; an audit log entry is created (FR-23); the graph view updates.
- On Reject: entities remain separate; the rejection is logged with reason (optional free-text note).
- Empty state:** "No pending matches for this case" — with a subtle confirmation this is expected, not an error.

6.2 Explore the Connected Network

- From Case Detail, investigator opens the **Graph View**, defaulting to a 1-hop neighborhood around the case's primary entity.
- Investigator can click a node to open the **Entity Detail Panel** (shows attributes + provenance + confidence + "Expand connections" button).
- Investigator can filter the visible graph by entity type, date range, or confidence level via a filter bar above the canvas.
- Investigator can select two nodes and choose **"Find Path"** to jump into the Relationship Path Explorer.

6.3 Trace a Relationship Path

- From Graph View or a direct search, investigator selects Entity A and Entity B.
- System queries `/graph/path` and renders the path as a linear, ordered chain of entities and relationship labels.
- If no path exists: explicit **"No connection found between these entities"** message (per SRS EH-4) — never a blank screen.
- Each hop in the path shows its confidence level and provenance tag inline.

6.4 Review Audit Log (Supervisory Officer)

- Supervisor opens **Audit Log**, filters by case, user, or date range.
- Each entry shows: timestamp, user, action type (view/confirm/reject/query), and affected entity/case reference.
- Log is read-only — no edit or delete controls are rendered for any role (matches SRS FR-25 at the UI level, not just backend enforcement).

7. Core Components

Component	Description
Graph Canvas	Cytoscape.js-rendered network; nodes styled by entity type (icon) and confidence level (border style); edges styled by relationship type tag (solid = direct/source-backed, dashed = design/synthetic)
Confidence Badge	Small label + icon (not color alone) — "High / Medium / Low" — attached to entities, relationships, and analytic findings wherever shown
Provenance Tag	Shows <code>Source_Dataset</code> and <code>Provenance_Mode</code> in a

	consistent small-print format, visible in detail panels and on hover tooltips
Entity Card	Compact summary (used in lists/queues): entity type icon, name/identifier, confidence badge, provenance tag
Relationship Path Strip	Horizontal chain UI showing entities connected by labeled relationship arrows
Filter Bar	Entity type, date range, confidence level filters — applies to Graph View and Analytics screens
Timeline Component	Vertical or horizontal chronological list of events, each entity-linked
Status Chip	“Suggested” (dashed outline) vs “Confirmed” (solid fill) — used anywhere AI output appears, per BR-3
Data Table	Used for Audit Log, Case List, Entity Resolution Queue — sortable, filterable
Top Navigation Bar	Persistent role-gated navigation (Section 4)

8. Interaction Patterns

- **Node click** → opens Entity Detail Panel (side drawer, doesn't navigate away from graph).
- **Node hover** → tooltip showing name, type, and confidence badge.
- **Drag/pan/zoom** → standard Cytoscape.js canvas gestures.
- **Double-click a node** → expands its immediate connections (progressive disclosure, per Design Principle 4).
- **Multi-select two nodes** (shift-click) → surfaces a “Find Path” action button.
- **Filter bar changes** → graph re-renders in place, no full page reload.
- **Confirm/Reject buttons** → require a single click, immediate optimistic UI update, with a toast confirmation (“Match confirmed — entities merged”).

9. Forms & Validation

Form	Fields	Validation
Login	Username, Password	Required; error shown inline on failed auth (no field-specific error that reveals which field was wrong, per standard security practice)
Data Ingestion Upload	File (CSV/text), Source Dataset selector	File type restricted to accepted formats; Source Dataset is a required dropdown (feeds Source_Dataset metadata per SRS DR-7)
Entity Resolution Reject	Optional reason (free text)	No hard validation — optional field
Admin: Add User	Name, Email, Role	Email format validated; Role must be one of the 4 defined roles (dropdown, not free text)
Case/Entity Search	Search term, filters	Debounced input; empty search shows full list, not an error

10. Loading, Error, and Empty States

State	Behavior
Graph loading	Skeleton/placeholder graph shape with a subtle loading

	indicator — never a blank white canvas
Ingestion failure (SRS EH-1)	Non-blocking toast/banner: “X of Y records failed validation — [view details]”; successful records still process
No path found (SRS EH-4)	Explicit message in the Relationship Path Explorer, not an empty chain
No pending resolution matches	Friendly confirmation state, distinct from an error (“You’re all caught up”)
No search results	“No cases match your search” with a suggestion to adjust filters
Session expired	Redirect to Login with a preserved return-path and a clear “Session expired, please log in again” message
Confidence score uncomputable (SRS EH-5)	Entity/relationship rendered with a “Low confidence — flagged for review” state, never silently hidden or upgraded

11. Responsive Behavior

- **Primary target: desktop/laptop workstation** (1280px+) — investigators work at fixed workstations; this is the design priority.
- **Tablet (768-1024px):** supported for Supervisory Officer review flows (Audit Log, Case Summaries, Analytics) — graph view remains usable but with a simplified default zoom level.
- **Mobile:** explicitly out of scope for MVP (per PRD Out-of-Scope) — the app should not break on a mobile viewport, but no dedicated mobile layout is designed. A minimal “best viewed on desktop” notice is acceptable if accessed on a small screen.

12. Accessibility

- **Target: WCAG 2.1 AA** where feasible for MVP.
- **Never encode meaning by color alone** — confidence levels and provenance always pair color with text/icon (Confidence Badge, Section 7) — critical given colorblind users and the high-stakes nature of the data.
- **Keyboard navigation:** all forms, buttons, and the Entity Resolution Queue must be fully keyboard-operable. The graph canvas itself (Cytoscape.js) is inherently mouse-centric — provide an alternative **list/table view** of the same entities and relationships for keyboard and screen-reader users (also doubles as a useful audit-friendly view).
- **Contrast ratios:** minimum 4.5:1 for body text, 3:1 for large text/icons, per WCAG AA.
- **Focus states:** visible focus outlines on all interactive elements — do not remove default focus rings without replacing them.

13. Typography

Use	Font	Notes
UI text (body, labels, navigation)	Inter (or system-ui fallback)	Clean, highly legible sans-serif, standard for data-dense professional tools
Headings	Inter, semi-bold/bold weights	Same family, weight differentiation only — avoids visual clutter of multiple typefaces
IDs / Audit References / Technical values	A monospace font (e.g., “JetBrains Mono” or system monospace)	Distinguishes machine-generated identifiers from human-readable text at a glance

Type scale (practical, 4-5 sizes, not a large arbitrary scale): - Heading 1: 24px / bold - Heading 2: 18px / semi-bold - Body: 14px / regular - Small/meta (provenance tags, timestamps): 12px / regular

14. Color System

Restrained, professional palette — avoids consumer-app brightness; uses color purposefully for status/semantic meaning only.

Purpose	Color Direction	Notes
Primary (navigation, primary actions)	Deep navy/blue	Matches the trust/authority tone already used in the pitch deck
Secondary/accent	Muted teal or slate	Supporting UI elements, secondary buttons
Confidence: High	Green (paired with a checkmark icon, never color alone)	
Confidence: Medium	Amber/orange (paired with a caution icon)	
Confidence: Low	Red-orange (paired with a flag icon) — reserved, not used decoratively elsewhere	Avoid overusing red so it retains urgency when it matters
Synthetic entity indicator	Dashed border / muted fill	Visual distinction independent of the confidence color system
Source-backed entity indicator	Solid border / full fill	
Neutral/background	Off-white / light gray	Reduces eye strain for extended investigator use
Error states	Standard red, used only for true errors (not for Low confidence, to avoid over-alerting)	

Principle: confidence-level color and error-state color must be visually distinguishable from each other — a “Low confidence” entity is not an “error,” and conflating the two would train users to ignore real errors.

15. Spacing & Layout

- **Base unit:** 8px grid — all padding/margin values are multiples of 8 (4px allowed for tight inline spacing only).
 - **Standard content padding:** 24px around major panels/cards.
 - **Component gap (lists, form fields):** 16px.
 - **Graph canvas:** full available viewport height within its container, with a fixed-height filter bar above (56px) and an optional side drawer (360px width) for the Entity Detail Panel that overlays without resizing the canvas.
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16. Component States

Every interactive component must define, at minimum: - **Default** - **Hover** - **Active/Pressed** - **Disabled** (e.g., Confirm/Reject buttons disabled after one is clicked, to prevent double-submission) - **Focus** (visible outline, for accessibility)

Graph nodes additionally require: - **Default**, **Hovered**, **Selected**, **Highlighted** (e.g., nodes along a found relationship path are highlighted distinctly from the rest of the graph).

17. Implementation Notes (for the Development Plan)

- Use CSS variables (or a design-token file) for the color system (Section 14) so confidence/provenance styling stays consistent across the Graph Canvas (Cytoscape.js styling config), Entity Cards, and Badges without duplicated hex values.
- Cytoscape.js node/edge styles should be driven directly by the same `Confidence_Level` and `Synthetic_Flag` fields returned by the backend (Architecture Section 6.1) — no separate frontend-side confidence logic.
- The keyboard-accessible list/table alternative to the graph (Section 12) can reuse the same Data Table component defined for Audit Log and Case List, minimizing new component work.

This UI/UX Document is derived from the TraceX PRD v1.1, SRS v1.0, and Architecture v1.0. Screen names and roles are kept identical across all four documents to avoid drift.